

CHARLES (CHUCK) FORSYTH

Research Computing and AI Executive | HPC | Research Cloud | Navy Veteran

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EXECUTIVE SUMMARY

Research computing and AI executive with 30+ years in technology, from Navy radar to high-performance computing to generative AI. As Director of Research Computing at UC Riverside, I lead a matrix team of about 20 across ITS and the colleges and run the cloud, cluster, storage, security and AI services behind about 1,000 researchers in 22 colleges and schools, and I still build the platforms myself, now with a team of AI agents. I led UCR's first-of-its-kind research partnership with Google Cloud, served as Co-PI on a \$498K NSF CC* research storage award, and chaired the UC Research IT Committee across every UC campus. I have been building with AI since the GPT-2 era, running GPU jobs for it on the National Research Platform in 2020, and shipped an LLM assistant for HPC users in 2023. The number I manage to is time to science.

CAREER HIGHLIGHTS

- **Google Cloud partnership:** led Ursa Major, announced by Google in 2023 as a first-of-its-kind research subscription; 170+ research projects under one governed, budgeted cloud.
- **NSF award:** Co-PI on CC* award 2346636 (\$498,553), which delivered 2.09 PB of research storage on the 100 Gbps Science DMZ, 422 TB shared with the National Research Platform.
- **AI for research:** built UCR's governed AI stack (MCP servers, AI Gateway, elastic GPU cluster) and ship it with AI agents: 240+ releases across ten tools in five weeks.
- **Campus and systemwide leadership:** founded and lead UCR Research IT Leaders; Research Advisory Board and Academic Senate research committee; Chair of the UC Research IT Committee (2022-2024), uniting research IT leaders from every UC campus.

CORE COMPETENCIES

- **Leadership:** AI and research computing strategy, governance and policy; budgets and FinOps; NSF grants; vendor and industry partnerships; systemwide committees
- **AI:** agentic AI and MCP servers, governed model gateways (Gemini, Claude, open-weight), LLM tools for researchers, AI security and audit
- **Infrastructure:** HPC and Slurm/PBS, Google Cloud and AWS, Kubernetes, Ceph, Lustre, GPFS, InfiniBand, Linux; Python, Go, Bash; Postgres and pgvector
- **Security:** NIST 800-171, CMMC, P3/P4 data security plans, secure research enclaves

PROFESSIONAL EXPERIENCE

Director of Research Computing

Mar 2023 - Present

University of California, Riverside, Riverside, CA

Lead research computing strategy, services and AI for about 1,000 researchers in 22 colleges and schools through a matrix team of about 20: direct reports and student staff, the ITS systems team (dotted line for research) and research IT staff in colleges, centers and units across campus.

- Run Ursa Major, the Google Cloud research partnership: per-project budgets, automated spend throttles, security baselines and FinOps across 170+ projects.
- Wrote NSF CC* proposal 2346636 and its budget (Co-PI); designed CephRDS (2.09 PB usable) and its cost recovery model; six science driver labs migrated, 422 TB federated to the National Research Platform.
- Built the AI-for-research stack: MCP servers with per-user Google sign-in, roles and a full audit trail; the UCR AI Gateway; and an elastic Slurm cluster on Google Cloud (CPU, high-memory, NVIDIA L4 GPU) with zero compute cost when idle.
- Built Nexus, the system of record Research Computing runs on: 943 researchers, 431 org units, 176 cloud projects, 12,000+ links.
- Data security plans and secure enclaves for P3/P4 and CMMC research with the ISO.
- Founded and lead UCR Research IT Leaders, which coordinates research IT across campus; serve on the Research Advisory Board and the Academic Senate research committee.
- Official UCR PI status (2024); grant strategy; ship software with a team of AI agents under written specs, test gates and human approval.

Associate Director of Research Computing

Aug 2019 - Mar 2023

University of California, Riverside, Riverside, CA

- Developed research computing grant strategy and managed budgets for research technology initiatives within Information Technology Solutions (ITS).

- Lead systems architect for research computation and the campus Science DMZ; proxy for the CIO and CTO with the academic senate; supervised technical staff.
- Brought UCR researchers onto National Research Platform GPUs: TensorFlow, PyTorch and GPT-2 jobs (2019-2020).

Chair, UC Research IT Committee

Jul 2022 - Jan 2024

University of California (systemwide), Remote

Vice Chair, Jun 2021 - Jul 2022.

- Led the UC Research IT Committee, liaising with the UC IT Leadership Council (ITLC) and the UC Office of the President CIO.
- United research IT leaders from every UC campus around shared priorities; set and met annual goals and advocated systemwide for UC researchers' computing needs.

Lead Linux & HPC Systems Administrator / Consultant

Jun 2017 - Aug 2019

UC Riverside, High Performance Computing Center (HPCC), Riverside, CA

- Rebuilt the HPCC campus cluster and the server room it runs in, including new power and cooling; moved configuration management to Ansible and set up PXE bare-metal provisioning.
- Built the HPCC Cloud Service (2017-2018): lab-private HPC clusters in AWS created from the HPCC cluster with `hpcc_cloud`, a `cfnCluster` wrapper I wrote; 130 commits to the cluster's software modules.
- Taught 19 HPCC workshops (2017-2019; sole instructor on 10) and wrote the four training decks.
- XSEDE Campus Champion connecting UCR to NSF supercomputing centers; UCR representative to the UC Research IT Committee and CaRCC.

Lead Linux & HPC Systems Administrator / Research Consultant

Jan 2012 - Jun 2017

Corning Incorporated, Corning, NY

Lead HPC architect and administrator for Corning's research, development and engineering computing.

- Linux clusters with thousands of cores, Lustre, InfiniBand, SAN and NAS storage; 60+ network-licensed science codes (Ansys, COMSOL, Abaqus).
- 400+ hardware support contracts and 150 kVA+ of data center power and cooling; led internal and external projects and wrote training and documentation.

Earlier Career

- **Owner and Research System Administrator, CnL Networks**, Jun 2003 - Jan 2012. Founded and ran a systems administration and IT outsourcing business; concurrently Technical Account Manager, STCR (Apr 2008 - Aug 2009).
- **Linux System Administrator, Hancock Telephone Company**, Mar 1998 - Jun 2003.
- **Hardware Test Engineer, IBM**, Aug 1997 - Mar 1998; Technical Support, AOL, Mar - Aug 1997.
- **Advanced Electronics Technician, United States Navy**, Dec 1994 - Mar 1997. Radar, communications and IFF systems aboard the Aegis cruiser USS Lake Champlain (CG-57).

AI AND RESEARCH PLATFORMS I BUILT

- **ursa-bifrost**: MCP server that lets any AI assistant work a Slurm cluster through typed, audited tools (Go, MIT, github.com/UCR-Research-Computing/ursa-bifrost).
- **UCR AI Gateway**: one governed endpoint to Gemini, Claude and open models with per-lab budgets and a FinOps dashboard.
- **Nexus**: research relationship knowledge graph and OAuth MCP server with 41 role-gated tools.
- **deep-research**: Gemini Deep Research workstation that checks report claims as real HPC jobs (MIT, github.com/charles-forsyth/deep-research).
- **Ultra AI Workstation**: AI operator desk for mail, tickets, Slack and the cluster, with two-step approval on AI drafts (MIT).
- **Skywalker**: audit engine for hundreds of Google Cloud projects.
- **Ursa Major Ask (2023), uma (2024), HPC Job Gallery (2026)**: early LLM tools for HPC users; 49 tested Slurm workflows.

PROFESSIONAL SERVICE AND LEADERSHIP

- UCR Campus Champion, NSF XSEDE and ACCESS (2017-present); NSF grant peer reviewer (2020-2021)
- Co-Chair, UCR Visualization Lab Steering Committee; UCR representative to CaRCC, CASC and ACI-REF; research collaborator, Pacific Research Platform (2017-present)

PUBLICATIONS AND TEACHING

- "One Day on the Research Stack," Field Notes, Oct 2026. charles-forsyth.github.io/papers/
- The AI + HPC Field Manual, 15 chapters, 2026. charles-forsyth.github.io/handbook/
- Taught 19 HPCC workshops at UCR (2017-2019): Linux, cluster, Python and AWS HPC; NRP Nautilus researcher guide, 8 articles (2026)

EDUCATION AND CERTIFICATIONS

- B.S., Information Technology, Purdue University Global, 2015 (GPA 3.89); Networking and Technical Support Specialist, Ridley-Lowell Institute, 2011 (GPA 4.0); Advanced Electronics Technology, U.S. Navy, 1997.
- **Certifications**: PBS Professional (Altair), Mellanox InfiniBand, EMC Isilon, Kubernetes, PMI Project Management Foundations.